Portugal's Public-Debt Interest Burden and Debt Dynamics in the Euro Area

Diogo Ribeiro[1,2,*] ORCID

Department of Data Science and Research, MySense, London, UNITED KINGDOM.
School of Media Arts and Design, Polytechnic of Porto
PORTUGAL.
[*] *Corresponding Author: dfr@esmad.ipp.pt*

July 2026

Abstract

This paper provides a reproducible empirical assessment of Portugal's general government interest burden from 1995 to 2025, the harmonised Eurostat ESA 2010 period used for the main analytical sample. The principal measure is interest payable by the general government sector as a percentage of nominal GDP. The analysis separates three quantities that are frequently conflated in fiscal commentary: the national-accounts interest expense, the interest burden relative to GDP, and the market yield on new benchmark debt. It combines source-provenance review, national-accounts measurement, descriptive periodisation, discrete debt-dynamics accounting, cross-country Eurostat comparison, and deterministic refinancing simulations. The central empirical result is that Portugal's interest burden fell from 5.50 percent of GDP in 1995 to 1.90 percent of GDP in 2025, despite a higher debt ratio in 2025 than in the mid-1990s and despite a material nominal interest bill of EUR 5.96 billion. In the exact 2014–2025 decomposition, the reconstructed burden changed by -2.87 percentage points, with -1.64 percentage points from the average financing-cost effect and -1.22 percentage points from the debt-exposure effect. By 2025, Portugal ranked sixth among the twenty euro-area country comparators by interest expenditure as a percentage of GDP, below Italy, Greece, Spain, Belgium, and France. The paper emphasises that rate shocks pass through gradually to the average cost of the outstanding stock; the static full-pass-through effect of a 100 basis-point shock at the 2025 debt ratio is approximately 0.90 percentage points of GDP, but this is an arithmetic long-run sensitivity rather than a one-year forecast.

Keywords: Portugal; public debt; interest expenditure; fiscal sustainability; debt dynamics; Eurostat; ESA 2010; euro area.
JEL codes: E62; H62; H63; H68.

Contents

1 Introduction

Interest expenditure is the fiscal price of previously accumulated public liabilities. It is not merely an accounting residual: it constrains the composition of public expenditure, affects the primary balance required to stabilise debt, and mediates the transmission of monetary and sovereign-risk conditions to the public budget. In high-debt economies, a modest movement in the effective interest rate can have large fiscal implications when compounded over the debt stock. Conversely, a high nominal interest bill may coexist with a falling interest burden when nominal GDP grows or when the debt ratio falls.

Portugal is a useful case for studying this interaction. The country entered monetary union after a period of declining long-term yields, experienced a large rise in debt and financing stress during the global financial crisis and the euro-area sovereign-debt crisis, then benefited from lower effective financing costs and later from rapid post-pandemic nominal GDP growth. These episodes make it possible to observe several regimes in one coherent annual dataset: euro convergence, early monetary union, crisis, adjustment, the ECB intervention and refinancing period, the pandemic, and the recent inflation and monetary-tightening regime.

The purpose of this paper is not to estimate a structural fiscal reaction function or to forecast Portugal's debt path. Its purpose is narrower and more foundational: to measure the interest burden correctly, document its evolution, and show how the burden should be interpreted in the presence of changing GDP, debt ratios, effective interest rates, and refinancing lags. The empirical strategy is therefore accounting-based and transparent. It treats national accounts as the measurement system, then uses deterministic identities and carefully labelled comparisons to organise the evidence.

The paper answers four research questions:

1. How have the euro amount of Portugal's general-government interest and the GDP-normalised burden evolved since 1995, and why do the two move differently?

2. How much of the change in the burden is attributable to the average financing cost and how much to debt exposure?

3. Where does Portugal sit in the euro-area distribution of interest burdens under harmonised definitions?

4. Under explicit assumptions, how quickly does a benchmark rate shock reach the interest bill through refinancing?

The central finding is that Portugal's public-debt interest burden declined substantially over the harmonised Eurostat period. The burden was 5.50 percent of GDP in 1995, 4.80 percent in 2014, 2.90 percent in 2019, and 1.90 percent in 2025. This does not mean that interest payments became fiscally irrelevant. In 2025 the general government still paid EUR 5.96 billion in interest. Rather, the result means that the economy's nominal scale and the effective cost of the outstanding debt stock changed enough to reduce the interest-to-GDP ratio.

2 Conceptual Framework and Measurement

Public discussion often uses the word "interest" to refer to several different objects. This paper distinguishes them explicitly.

First, *interest payable* is a national-accounts flow. In the Eurostat ESA framework it is the interest expense recorded for the general-government sector, not only the central government and not only cash paid in a particular budget line. This is the numerator used in the headline outcome and follows the ESA national-accounts measurement framework [1, 2].

Second, the *interest burden* is interest payable divided by nominal GDP. This is the main principal measure because it expresses the fiscal flow in relation to the annual income base. The ratio is appropriate for fiscal-space questions because the same nominal payment can be more or less burdensome depending on nominal GDP.

Third, the *market yield* is a price on new or benchmark borrowing. It is not the average cost of the outstanding debt portfolio. Because governments roll over debt gradually, benchmark yields affect the interest bill with a lag whose length depends on maturity, issuance strategy, cash buffers, and the composition of fixed-rate and variable-rate liabilities.

The empirical design of the paper follows from this distinction. The baseline outcome is not the bond yield, and the yield is not treated as if it repriced the entire stock immediately. Instead, the paper measures the national-accounts interest burden and separately analyses the gap between the benchmark yield and the average-debt rate.

Let D_t denote the nominal stock of general-government gross debt at the end of year t, Y_t nominal GDP, I_t interest payable, and PB_t the primary balance. Define $d_t = D_t/Y_t$, $i_t = I_t/Y_t$, and $pb_t = PB_t/Y_t$. A standard discrete debt-dynamics identity writes the change in the debt ratio as:

$$\Delta d_t = \frac{r_t^{DD} - g_t}{1 + g_t} d_{t-1} - pb_t + sfa_t, \tag{1}$$

where $r_t^{DD} = I_t/D_{t-1}$ is the debt-dynamics interest rate, g_t is nominal GDP growth, and sfa_t is the stock-flow adjustment. The first term captures the automatic debt dynamics generated by the difference between the debt-dynamics rate and nominal growth. A positive primary balance reduces the debt ratio, while a positive stock-flow adjustment increases it.

Equation 1 is not a behavioural model. It is an accounting identity once sfa_t is defined as the reconciliation residual. Its value in this paper is diagnostic: it separates periods in which debt pressure comes from the interest-growth differential from periods in which the primary balance or residual stock-flow movements dominate the observed debt change. This accounting tradition is standard in fiscal-sustainability analysis [3, 4].

Ignoring stock-flow adjustments, the primary balance that stabilises the debt ratio is approximately:

$$pb_t^* = \frac{r_t^{DD} - g_t}{1 + g_t} d_{t-1}. \tag{2}$$

When nominal growth exceeds the debt-dynamics interest rate, this term can be negative: the debt ratio may fall even with a primary deficit, provided the deficit is not too large and residual stock-flow movements do not offset the effect. When the debt-dynamics interest rate exceeds nominal growth, a primary surplus is required to stabilise the debt ratio absent stock-flow adjustments.

This stabilisation condition is crucial for interpreting Portugal after 2021. High nominal growth pushed the interest-growth differential into strongly negative territory in 2022 and 2023, lowering the debt-stabilising primary balance. That favourable arithmetic did not eliminate fiscal risk, but it did reduce the near-term debt-ratio pressure from the existing stock.

3 Data and Research Design

The unit of observation is the calendar year. The institutional sector is general government as defined in the European national-accounts framework. This sector choice is important because central-government cash measures can differ from general-government accrual-based national accounts. The paper therefore does not mix central-government budget execution with ESA general-government interest payable.

The main analytical sample is 1995–2025, the period in which the authoritative Portugal series is built from Eurostat ESA 2010 concepts. Every figure, table, decomposition, and conclusion in this paper uses that sample. The first ESA 2010 main-series year is marked as a basis break.

The core evidence is drawn from Eurostat government-finance statistics, government-debt statistics, national accounts, and the harmonised long-term government bond yield series [2]. These sources provide the main empirical objects required by the paper: interest payable, total government expenditure, gross public debt, nominal and real GDP, and a market-yield benchmark. The analysis uses million-euro values when the level of the interest bill or total expenditure is the object of interest and percentage-of-GDP measures when the fiscal burden or fiscal scale is the object of interest.

The European comparison is constructed under the same conceptual definitions: general government, annual frequency, harmonised national-accounts concepts, and comparable units. Aggregate geographies, such as the euro area, are retained as contextual information where available but excluded from country ranks. This design prevents the country comparison from being driven by inconsistent accounting definitions.

Forecast observations are excluded from historical descriptive statements unless explicitly stated.

This conservative treatment is appropriate because a long time series can create a false appearance of precision when accounting regimes change. The paper therefore uses the extension as context and the Eurostat ESA 2010 sample as the primary evidence base.

The principal burden measure is:

$$b_t = \frac{I_t}{Y_t} \times 100, \tag{3}$$

where I_t is general-government interest payable and Y_t is nominal GDP. The preferred empirical measure is Eurostat's official percentage-of-GDP ratio. A reconstructed ratio from the underlying million-euro values is used only as a reconciliation check and fallback.

The average-debt interest rate used for descriptive average-cost analysis is:

$$r_t^{AVG} = \frac{I_t}{(D_{t-1} + D_t)/2} \times 100. \tag{4}$$

The denominator is the average of prior-year and current-year debt. This choice reduces mechanical distortion when year-end debt moves sharply. It also matches the paper's emphasis on the average cost of the outstanding stock rather than the marginal cost of new borrowing.

The interest-burden decomposition uses the exact factorisation $b_t^R = r_t^{AVG} d_t^{AVG}$, where $d_t^{AVG} = ((D_{t-1} + D_t)/2)/Y_t$ and b_t^R is the reconstructed burden from unrounded nominal interest and GDP. Between endpoint years 0 and 1, the symmetric exact decomposition is:

$$b_1^R - b_0^R = \frac{d_1^{AVG} + d_0^{AVG}}{2} \left(r_1^{AVG} - r_0^{AVG} \right) + \frac{r_1^{AVG} + r_0^{AVG}}{2} \left(d_1^{AVG} - d_0^{AVG} \right). \tag{5}$$

The first component is the average financing-cost effect. The second is the debt-exposure effect. No separate interaction term is used in the principal decomposition because the symmetric formula allocates the full endpoint change exactly to the two reported components.

Interest expenditure is recorded as a positive expense. Therefore:

$$pb_t = ob_t + b_t, \tag{6}$$

where ob_t is the overall balance as a percentage of GDP. A government can therefore have an overall deficit and a primary surplus if the deficit is smaller than interest expenditure.

Nominal GDP growth is calculated from current-price GDP. Real GDP growth is read from the Eurostat chain-linked volume percentage-change series. GDP deflator growth is derived as:

$$1 + \pi_t^{GDP} = \frac{1 + g_t^{nominal}}{1 + g_t^{real}}. \tag{7}$$

This multiplicative definition is preferable to subtracting real growth from nominal growth when growth rates are large, as in 2020–2023.

The full-pass-through interest-burden effect of a rate shock is:

$$\Delta b = d\Delta r. \tag{8}$$

At a debt ratio of 89.70 percent of GDP, a 100 basis-point rate shock applied to the full stock implies an additional burden of approximately 0.897 percentage points of GDP. This is a static sensitivity. It should not be interpreted as an immediate forecast because only a fraction of the debt stock is refinanced in a typical year.

4 Data Quality and Validation

The empirical series passed every error-severity data-validation check (0 failures). 1 warning-severity check did not pass: the published debt ratio and the ratio reconstructed from levels differ by more than the configured tolerance of 0.15 percentage points in 1997, 1998.

The exact figures are these. In 1997 the published ratio is 58.70 percent of GDP against a reconstructed 57.6848 percent, a difference of 1.0152 percentage points. In 1998 the published ratio is 55.60 percent against a reconstructed 55.9522 percent, a difference of -0.3522 percentage points. The largest absolute discrepancy anywhere in the sample is 1.0152 percentage points. Both breaches are warning-level, not error-level.

The two figures are drawn from different Eurostat datasets: the published ratio from the excessive-deficit-procedure table and the reconstruction from the national-accounts GDP series. Nothing in the data identifies which side moved, so the cause is recorded as Unknown rather than guessed. Per-year detail is written to `reports/validation_detail.csv` and `reports/validation_detail.md`.

The analysis depends on three forms of data integrity. First, every observation must represent one and only one calendar year. Second, denominators such as GDP and debt must be positive, finite, and economically interpretable. Third, cross-country comparisons must use the same institutional sector, transaction concept, and unit. The validation layer checks these conditions before the figures and tables in the paper are produced. Technical replication details are reported in the appendices rather than in the main argument.

5 Portuguese Interest Burden, 1995–2025

The harmonised ESA 2010 sample contains 31 annual observations from 1995 to 2025. Table 1 summarises the main variables. Over this period, interest expenditure averaged 3.29 percent of GDP, with a minimum of 1.90 percent and a maximum of 5.50 percent. The nominal interest bill averaged EUR 5.53 billion, with a maximum of EUR 8.33 billion in 2014. The debt ratio averaged 92.26 percent of GDP, with a minimum of 54.20 percent in 2000 and a maximum of 134.10 percent in 2020.

Table 1: Summary statistics, Portugal ESA 2010 sample, 1995–2025

Variable	N	Mean	Std. dev.	Min	Min year(s)	Max	Max year(s)
Interest/GDP (%)	31	3.287	0.969	1.900	2022, 2025	5.500	1995
Interest (EUR m)	31	5531.365	1535.291	3475.400	1998	8334.400	2014
Government expenditure/GDP (%)	31	45.558	3.056	42.000	2023	51.900	2010
Government expenditure (EUR m)	31	80221.719	23093.810	38798.300	1995	130941.400	2025
Government revenue/GDP (%)	31	41.526	2.023	37.400	1995	44.700	2013
Government revenue (EUR m)	31	73889.542	24391.219	34084.900	1995	133000.000	2025
Debt/GDP (%)	31	92.258	29.324	54.200	2000	134.100	2020
Average-debt rate (%)	30	3.849	1.470	1.708	2022	7.934	1996
Overall balance/GDP (%)	31	-4.035	2.957	-11.400	2010	1.100	2023
Primary balance/GDP (%)	31	-0.748	2.725	-8.500	2010	3.200	2023
Nominal GDP growth (%)	30	4.211	4.063	-6.274	2020	12.686	2022
Real GDP growth (%)	30	1.527	3.021	-8.200	2020	7.000	2022
GDP deflator growth (%)	30	2.625	1.625	-0.325	2012	7.488	2023
Ten-year yield (%)	31	4.524	2.718	0.300	2021	11.470	1995

Notes: Observed Eurostat ESA 2010 Portugal rows only. The average-debt rate uses average debt. Interest is general-government interest payable. Where an extreme is attained in more than one year, every year is listed.

The most important descriptive fact is the decline in the interest burden. In 1995, Portugal's general government paid interest equal to 5.50 percent of GDP. In 2025 the burden was 1.90 percent of GDP. The change is a reduction of 3.60 percentage points of GDP. In nominal terms, however, the interest bill increased from EUR 5.04 billion to EUR 5.96 billion. This divergence is not a contradiction. It shows why the GDP denominator is essential for fiscal interpretation.

Figure 1 shows the long decline, the crisis-period reversal, and the renewed compression after 2014. The burden fell sharply from the mid-1990s into the early euro period, then rose during the sovereign-debt crisis. It declined again during the low-rate and asset-purchase period and remained near 2 percent of GDP in the inflation and monetary-tightening period.

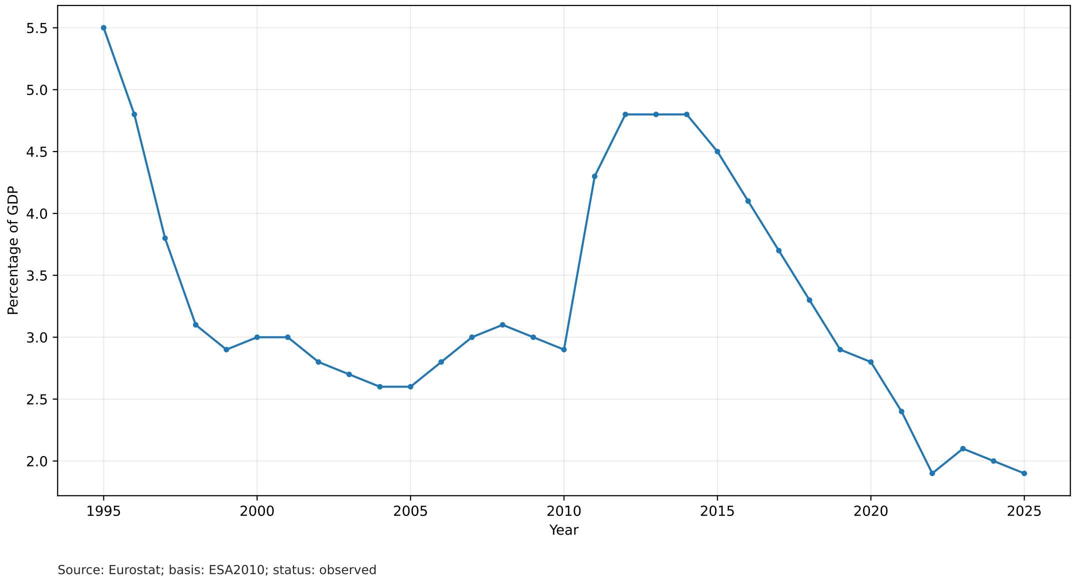


Figure 1: Portugal: interest expenditure as a percentage of GDP.

Figure 2 displays the same interest flow in nominal euros. The euro amount peaked in 2014 at EUR 8.33 billion and then declined through 2022 before increasing again in 2023–2025. The contrast between Figures 1 and 2 is central: a rising nominal bill can coexist with a stable or falling GDP burden when nominal GDP rises rapidly or when the debt ratio falls.

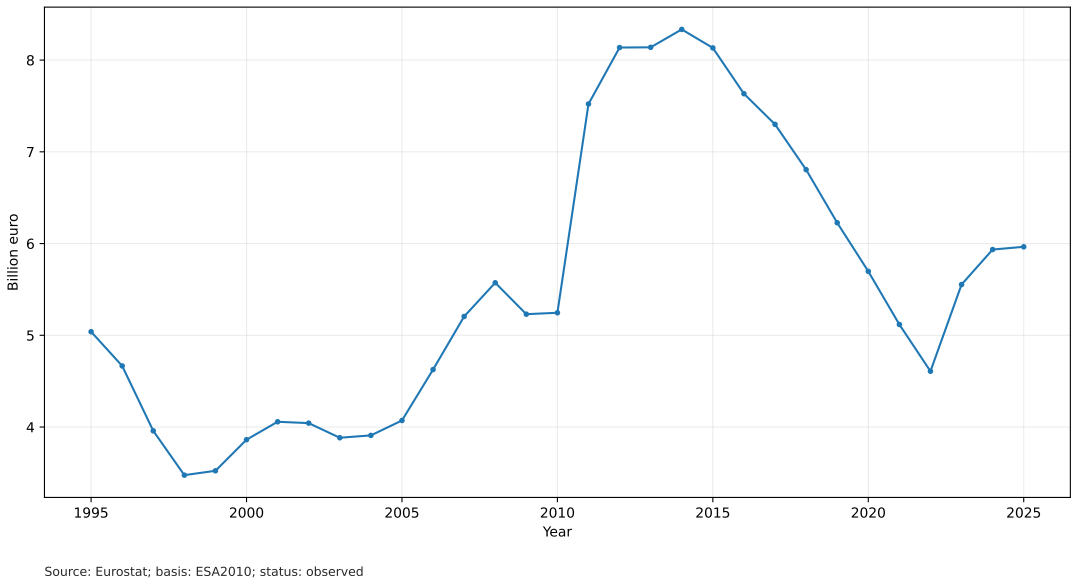


Figure 2: Portugal: interest expenditure in billion euros.

Table 2 places the interest bill inside the fiscal envelope for selected years. Percent of GDP is the right denominator for comparison across countries and over time, but it is not the denominator a budget works in: the share of total expenditure answers the more concrete question of how much of what the state spends buys nothing current. Revenue is shown alongside so the envelope within which interest is financed is visible.

Table 2: Interest against the fiscal envelope, selected years

Year	Interest (% of GDP)	Expenditure (% of GDP)	Interest (% of expenditure)	Revenue (% of GDP)	Interest (% of revenue)
1995	5.50	42.60	12.91	37.40	14.71
2012	4.80	48.80	9.84	42.60	11.27
2020	2.80	49.10	5.70	43.40	6.45
2025	1.90	42.70	4.45	43.40	4.38

Notes: Percent of GDP is the right denominator for comparison across countries and time; the share of total expenditure is the denominator a budget works in. Source: author calculations from Eurostat data.

Table 3: Regime averages, Portugal ESA 2010 sample

Regime	Years	Int./GDP	Int. EUR m	Debt/GDP	Avg. debt rate	Nom. growth	Prim. bal.
Euro convergence	1995–1998	4.30	4284.68	59.95	6.74	6.74	-0.20
Early euro period	1999–2007	2.82	4131.03	64.07	4.64	5.27	-1.54
Global financial crisis	2008–2010	3.00	5349.47	87.67	3.67	0.85	-5.37
Sovereign-debt crisis and adjustment	2011–2014	4.67	8033.05	126.47	3.81	-0.91	-1.95
Low-rate and asset-purchase period	2015–2019	3.70	7219.64	125.08	2.98	4.37	1.76
Pandemic	2020–2021	2.60	5407.45	129.00	2.05	0.71	-1.70
Inflation and monetary tightening	2022–2025	1.98	5515.15	97.83	2.05	9.14	2.50

Notes: All entries are regime means. Interest in euros is reported in million euros. The ten-year yield is excluded from the printed table to keep the regime table readable; its regime means are discussed in text.

The convergence period combines a high interest burden with a declining effective rate. The mean interest burden was 4.30 percent of GDP, and the mean average-debt interest rate was 6.74 percent. The ten-year yield averaged 7.82 percent. The debt ratio was relatively modest by later Portuguese standards, averaging 59.95 percent of GDP, but the high effective rate made interest expenditure large relative to GDP.

The early euro period had a lower interest burden, averaging 2.82 percent of GDP. The average-debt rate fell to a mean of 4.64 percent, while nominal growth averaged 5.27 percent. The debt ratio increased from the low point of 2000 to 72.70 percent by 2007, but the interest burden remained near 3 percent of GDP. This is the first major example in the sample of debt and interest burden moving differently.

During 2008–2010, nominal growth slowed sharply and the primary balance deteriorated. The mean primary balance was -5.37 percent of GDP. The debt ratio rose rapidly, and the overall balance reached its sample minimum of -11.40 percent of GDP in 2010. The interest burden had not yet fully adjusted to later sovereign stress, remaining around 3 percent of GDP on average.

The sovereign-debt crisis and adjustment period is the highest-burden regime in the harmonised sample. Interest expenditure averaged 4.67 percent of GDP and EUR 8.03 billion. The debt ratio averaged 126.47 percent of GDP. Nominal growth was negative on average, and the interest-growth differential was therefore unfavourable. The maximum nominal interest bill, EUR 8.33 billion, occurred in 2014.

Between 2015 and 2019, the interest burden declined from 4.50 percent to 2.90 percent of GDP. The average-debt interest rate declined from 3.50 percent to 2.50 percent. Primary balances became positive on average. Debt remained high, but the refinancing environment and effective-rate compression reduced the budgetary burden.

The pandemic period illustrates denominator and stock effects. Real GDP fell 8.20 percent in 2020, nominal GDP fell 6.27 percent, and the debt ratio jumped to 134.10 percent of GDP.

Yet the interest burden did not rise; it was 2.80 percent in 2020 and 2.40 percent in 2021. The reason is that the average-debt interest rate continued to decline, reaching 1.90 percent in 2021.

The recent period combines higher benchmark yields with strong nominal growth and a declining debt ratio. The interest burden averaged 1.98 percent of GDP. The average-debt rate averaged 2.05 percent, while nominal GDP growth averaged 9.14 percent. The debt ratio fell from 111.20 percent in 2022 to 89.70 percent in 2025. This period is not risk-free: nominal interest payments rose by EUR 1.36 billion from 2022 to 2025. But the interest-to-GDP ratio stayed at approximately 2 percent because nominal GDP and the debt ratio moved favourably.

Figure 3 decomposes nominal GDP growth into real growth and GDP deflator growth. The recent period is dominated by strong nominal growth. In 2022 nominal GDP grew 12.69 percent, real GDP grew 7.00 percent, and GDP deflator growth was 5.31 percent. In 2023 nominal GDP grew 10.82 percent and deflator growth was 7.49 percent. These values materially reduced debt-ratio pressure because the denominator grew quickly.

Figure 3: Portugal: nominal growth, real growth, and GDP-deflator growth.

Figure 4 reports the accounting decomposition implied by Equation 1. The pandemic year 2020 is the clearest example of adverse debt dynamics: nominal GDP contracted sharply, pushing the interest-growth term to 10.61 percentage points of GDP. By contrast, 2022 and 2023 show strongly favourable interest-growth arithmetic. The debt-stabilising primary balance was -12.06 percent of GDP in 2022 and -8.80 percent in 2023, before stock-flow effects. Both quantities are evaluated on the debt-dynamics rate $r_t^{\mathrm{dd}} = I_t/D_{t-1}$ used in Equation 1, not on the average-debt descriptive rate of Equation 4.

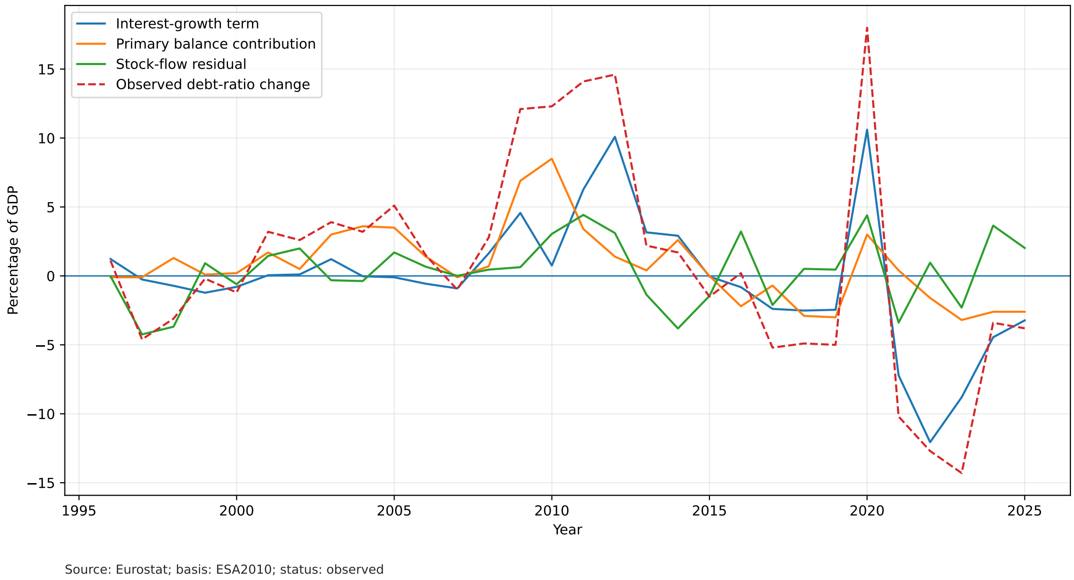


Figure 4: Portugal: debt-dynamics decomposition.

Table 4: Debt-dynamics diagnostic table, Portugal, 2020–2025

Year	Interest/GDP (%)	Lagged debt/GDP (%)	Nominal growth (%)	r^{AVG} (%)	r^{DD} (%)	Interest-growth (pp)	Primary balance (pp)	Stock-flow (pp)	Observed Δd (pp)	Rebuilt Δd (pp)	Error (pp)
2020	2.80	116.10	-6.27	2.20	2.29	10.61	3.00	4.39	18.00	18.00	$< 10^{-10}$
2021	2.40	134.10	7.69	1.90	1.90	-7.21	0.40	-3.39	-10.20	-10.20	$< 10^{-10}$
2022	1.90	123.90	12.69	1.71	1.72	-12.06	-1.60	0.96	-12.70	-12.70	$< 10^{-10}$
2023	2.10	111.20	10.82	2.08	2.05	-8.80	-3.20	-2.30	-14.30	-14.30	$< 10^{-10}$
2024	2.00	96.90	7.19	2.23	2.27	-4.45	-2.60	3.65	-3.40	-3.40	$< 10^{-10}$
2025	1.90	93.50	5.85	2.18	2.20	-3.23	-2.60	2.03	-3.80	-3.80	$< 10^{-10}$

Notes: Every column is a percentage (%) or a percentage point (pp), as marked in the heading; no decimal ratios are displayed. r^{AVG} is the average-debt rate and r^{DD} the debt-dynamics rate. The first row uses the previous debt ratio available in the processed dataset. Source: author calculations from Eurostat data.

The stock-flow adjustment is a residual. It absorbs valuation effects, financial transactions, cash-debt timing differences, and any discrepancy between the simple debt-dynamics identity and the measured debt ratio. In this dataset, it ranges from -4.24 percentage points of GDP in 1997 to 4.43 percentage points in 2011. It should not be interpreted as an estimated policy instrument. It is an accounting reconciliation term. Across the sample the identity closes to within $< 10^{-9}$ percentage points, so the residual carries no arithmetic error.

Figure 5 applies Equation 5 to selected endpoint intervals. The principal decomposition has two components: the average financing-cost effect and the debt-exposure effect. For 2014–2025, the reconstructed interest burden changed by -2.87 percentage points of GDP. The average financing-cost effect contributed -1.64 percentage points, while the debt-exposure effect contributed -1.22 percentage points. The dominant component was rate, and the reconciliation error was $< 10^{-10}$ percentage points. For 1996–2025, the reconstructed burden changed by -2.88 percentage points. The average financing-cost effect contributed -4.31 percentage points, while the debt-exposure effect contributed 1.42 percentage points. The dominant component was rate, and the reconciliation error was $< 10^{-10}$ percentage points.

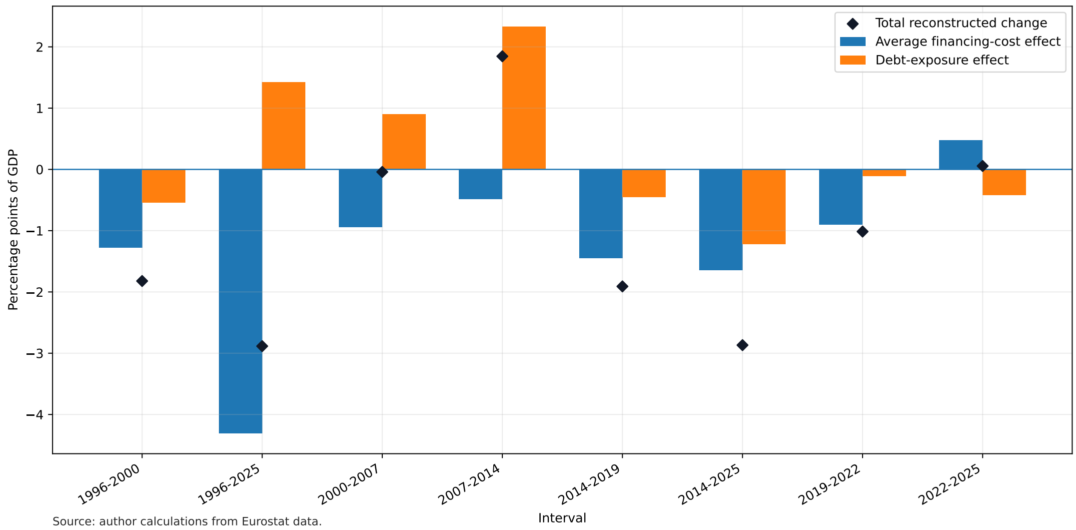


Figure 5: Portugal: endpoint decomposition of the reconstructed interest-burden change.

Table 5: Endpoint decomposition of reconstructed interest-burden changes

Start	End	Start burden	End burden	Total (pp)	Financing cost (pp)	Debt exposure (pp)	Error (pp)	Dominant	Official start	Official end
1996	2000	4.83	3.01	-1.822	-1.279	-0.543	$< 10^{-10}$	Financing cost	4.80	3.00
2000	2007	3.01	2.97	-0.041	-0.944	0.903	$< 10^{-10}$	Financing cost	3.00	3.00
2007	2014	2.97	4.81	1.846	-0.486	2.332	$< 10^{-10}$	Debt exposure	3.00	4.80
2014	2019	4.81	2.90	-1.909	-1.452	-0.457	$< 10^{-10}$	Financing cost	4.80	2.90
2019	2022	2.90	1.89	-1.014	-0.901	-0.113	$< 10^{-10}$	Financing cost	2.90	1.90
2022	2025	1.89	1.94	0.056	0.476	-0.420	$< 10^{-10}$	Financing cost	1.90	1.90
2014	2025	4.81	1.94	-2.868	-1.645	-1.223	$< 10^{-10}$	Financing cost	4.80	1.90
1996	2025	4.83	1.94	-2.885	-4.308	1.423	$< 10^{-10}$	Financing cost	4.80	1.90

Notes: Burdens are percent of GDP; effects and errors are percentage points. Effects are shown to three decimals so that the two components add to the displayed total. The decomposition uses unrounded nominal interest, debt, and GDP. Source: author calculations from Eurostat data.

Table 6 reports arithmetic counterfactuals for 2014 and 2025. These calculations hold one factor fixed while replacing the other with the endpoint value. They are accounting comparisons, not causal estimates.

Table 6: Arithmetic interest-burden counterfactuals, 2014 and 2025

Year	Scenario	Average-debt rate (%)	Debt exposure (% of GDP)	Burden (% of GDP)
2014	Observed	3.68	130.70	4.81
2025	Observed	2.18	88.99	1.94
2014	2014 rate with 2025 debt exposure	3.68	88.99	3.28
2025	2025 rate with 2014 debt exposure	2.18	130.70	2.86

Notes: These are arithmetic counterfactuals, not causal estimates. Every column is a percentage, as marked in the heading. Source: author calculations from Eurostat data.

Table 7 provides recent annual observations. The key pattern is visible in the last four rows. From 2022 to 2025 the nominal interest bill rose from EUR 4.61 billion to EUR 5.96 billion, but interest as a share of GDP remained at 1.90 percent in both endpoint years. Over the same period the debt ratio fell by 21.50 percentage points, while the average-debt interest rate rose by 0.48 percentage points.

Table 7: Recent annual dynamics, Portugal

Year	Int./GDP (%)	Interest (EUR m)	Debt/GDP (%)	Avg. debt rate (%)	Overall bal. (% GDP)	Primary bal. (% GDP)	Nom. growth (%)
2014	4.8	8334.4	132.5	3.68	-7.4	-2.6	1.47
2015	4.5	8132.8	131.0	3.50	-4.5	0.0	3.58
2016	4.1	7633.3	131.2	3.18	-1.9	2.2	3.90
2017	3.7	7299.4	126.0	2.97	-3.0	0.7	4.90
2018	3.3	6805.8	121.1	2.75	-0.4	2.9	4.85
2019	2.9	6226.9	116.1	2.50	0.1	3.0	4.63
2020	2.8	5697.1	134.1	2.20	-5.8	-3.0	-6.27
2021	2.4	5117.8	123.9	1.90	-2.8	-0.4	7.69
2022	1.9	4608.0	111.2	1.71	-0.3	1.6	12.69
2023	2.1	5553.3	96.9	2.08	1.1	3.2	10.82
2024	2.0	5934.8	93.5	2.23	0.6	2.6	7.19
2025	1.9	5964.5	89.7	2.18	0.7	2.6	5.85

6 European Comparison and Rate-Shock Exposure

The comparator panel includes the complete euro-area country universe at the configured endpoint year, with euro-area and EU aggregates retained only as context. The country ranking excludes aggregate geographies. Each country-year observation uses the same Eurostat concepts for interest, GDP, debt, balance, and real growth. This prevents a common error in fiscal comparison: mixing country-specific cash or budgetary definitions with national-accounts definitions.

Eligibility and ranking method. A rank without a denominator is not a finding, so both are stated. A geography enters the ranking only if it is a euro-area member, is not an aggregate, and carries every required series: interest in euro and as a share of GDP, the debt ratio, nominal GDP, and the average-debt rate. Eligibility, availability, observation status, and the reason for every exclusion are written to `data/processed/euro_area_eligibility.csv`.

Observation status is derived from Eurostat's per-series flags rather than asserted. This matters: the panel's own status column labels every row as observed, while Eurostat flags 10 of the geographies as provisional in 2025, Portugal among them. The accepted statuses are configured, and provisional values are admitted but disclosed.

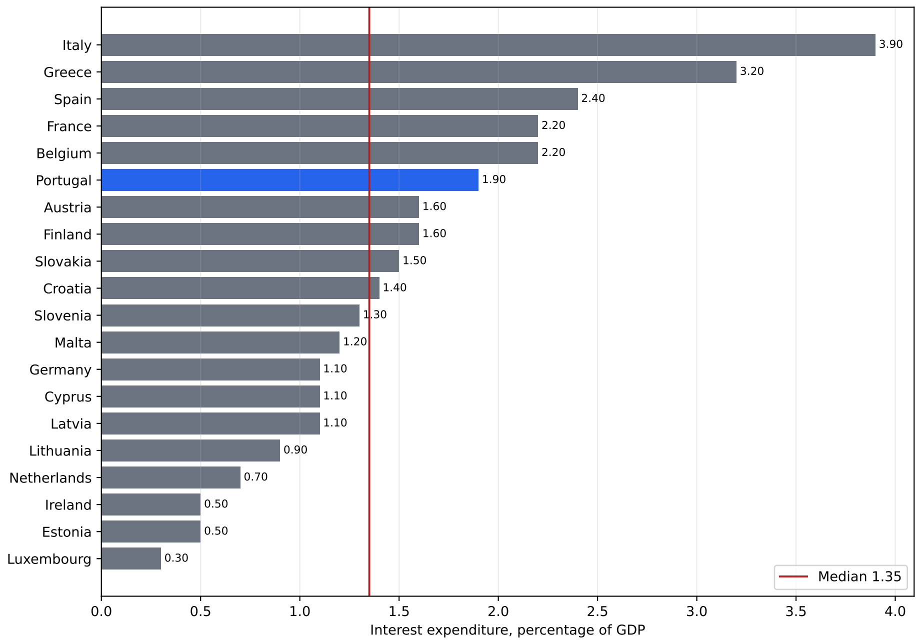
Ranking uses the competition convention, so tied values share the better rank and the next rank is skipped: 1, 2, 2, 4. The latest year in which every eligible country carries every required series is 2025.

Portugal's position. Table 8 reports the 2025 ranking by interest expenditure as a percentage of GDP. Portugal ranked 6 of 20 eligible euro-area countries, at the 75.0th percentile of the eligible distribution. The median burden was 1.35 percent of GDP, with an interquartile range from 1.05 to 1.98 percent; Portugal sat 0.55 percentage points above the median. 2 geographies were excluded (aggregate geography; not a euro-area member). Portugal's own 2025 observation status is provisional.

This is a comparison of the current interest burden. It is not a ranking of fiscal sustainability, which would require assumptions about growth, primary balances, and maturity structure that this table does not make.

Table 8: European comparator panel, 2025

Rank	Country	Interest/GDP (%)	Debt/GDP (%)	Avg. debt rate (%)	Ten-year yield (%)	Primary bal. (% GDP)
1	Italy	3.9	137.1	2.87	3.59	0.8
2	Greece	3.2	146.1	2.15	3.37	4.9
3	Spain	2.4	100.7	2.43	3.21	0.0
4	Belgium	2.2	107.9	2.14	3.17	-3.0
4	France	2.2	115.6	1.97	3.35	-2.9
6	Portugal	1.9	89.7	2.18	3.08	2.6
7	Austria	1.6	81.5	2.03	2.98	-2.6
7	Finland	1.6	88.5	1.92	3.00	-2.3
9	Slovakia	1.5	61.4	2.62	3.43	-3.0
10	Croatia	1.4	56.3	2.56	2.99	-1.6
11	Slovenia	1.3	65.7	1.98	3.10	-1.2
12	Malta	1.2	46.4	2.72	3.44	-1.0
13	Cyprus	1.1	55.0	1.99	2.93	4.5
13	Germany	1.1	63.5	1.79	2.59	-1.6
13	Latvia	1.1	46.9	2.44	3.27	-1.4
16	Lithuania	0.9	39.5	2.38	2.88	-0.9
17	Netherlands	0.7	44.4	1.67	2.80	-0.9
18	Estonia	0.5	24.1	2.08	3.31	-1.5
18	Ireland	0.5	32.9	1.40	2.89	2.3
20	Luxembourg	0.3	26.5	1.30	2.90	-1.7



Source: author calculations from Eurostat data. 20 eligible euro-area countries; aggregates excluded; 10 carry provisional Eurostat flags. Portugal highlighted.

Figure 6: European comparison of interest expenditure as a percentage of GDP. Eligible euro-area countries only, aggregates excluded; the vertical line marks the median. 10 geographies carry provisional Eurostat flags in 2025.

Portugal's intermediate position is consistent with three facts. First, Portugal's debt ratio remained high by northern euro-area standards but was below Italy, Greece, and Spain in 2025. Second, Portugal's average-debt interest rate was moderate relative to Spain and Italy. Third, Portugal ran a sizeable primary surplus in 2025, which supported debt reduction. The comparison therefore does not show that Portugal is low-risk in an absolute sense; it shows that its interest burden was below the highest-burden euro-area countries under harmonised definitions.

Table 9 reports static sensitivities at the 2025 debt ratio. A 50 basis-point full-stock shock implies 0.449 percentage points of GDP in additional annual interest. A 100 basis-point shock implies 0.897 percentage points. A 200 basis-point shock implies 1.794 percentage points. These numbers are useful for scale, but they deliberately overstate first-year budgetary effects when most debt is fixed-rate and matures gradually.

Table 9: Static full-pass-through sensitivities at 2025 debt ratio

Debt/GDP (%)	Shock (bps)	Shock rate decimal	Additional interest/GDP (percentage points)
89.70	50	0.005	0.449
89.70	100	0.010	0.897
89.70	200	0.020	1.794

The static bound above assumes the whole stock reprices at once. It does not. Only the part of the portfolio that matures is rolled over at the new rate, so the arithmetic effect of a shock arrives gradually. To make that explicit the paper uses a stylised cohort model.

The cohort recurrence. At the horizon origin the entire initial stock sits in a legacy cohort carrying the initial average portfolio rate r_0. In year t a configured share s_t of the *original* stock leaves the legacy cohort and is repriced at that year's new-issuance rate. Cohorts already repriced keep the rate they were assigned, so no cohort is repriced twice. Writing $S_t = \sum_{k \le t} s_k$ for the cumulative repriced share, the average portfolio rate is

$$\bar{r}_t = (1 - S_t)\, r_0 + \sum_{k \le t} s_k\, y_k, \tag{9}$$

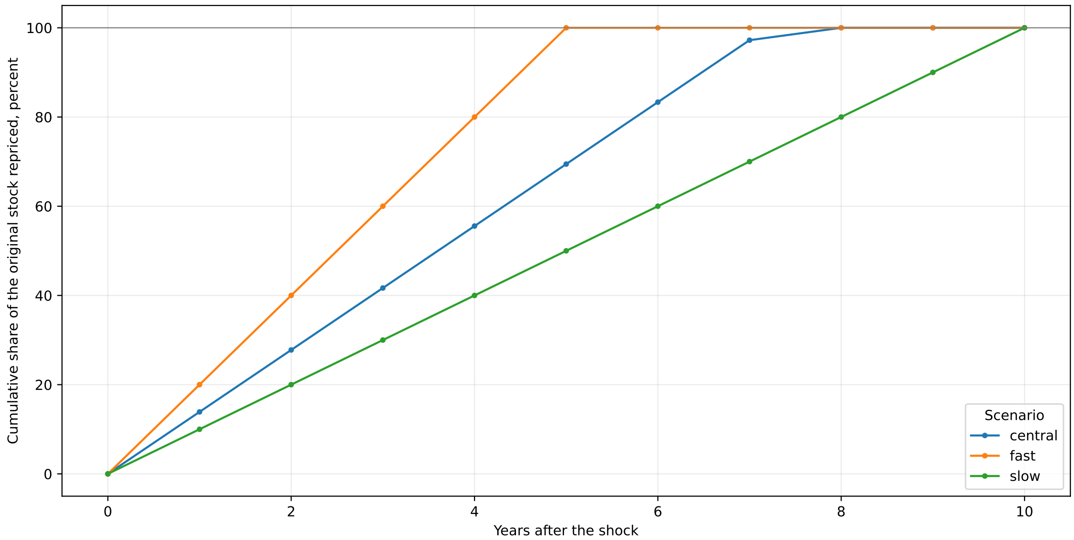
where y_k is the new-issuance rate in year k. The shocked path uses $y_k + \Delta$ and the baseline path uses y_k; the reported quantity is the difference between the two.

Table 10: Stylised refinancing scenario assumptions

Scenario	Annual repricing (%)	Implied maturity (years)	Horizon (years)	Initial portfolio rate (%)	Baseline issuance rate (%)	Debt (% of GDP)	Debt and GDP paths
Central (main)	13.89	7.2	10	2.18	3.08	89.70	fixed
Fast	20.00	5.0	10	2.18	3.08	89.70	fixed
Slow	10.00	10.0	10	2.18	3.08	89.70	fixed

Notes: A stylised cohort model, not a forecast. The central scenario applies a uniform annual repricing share consistent with the published average maturity of the debt stock of 7.2 years in 2024 (IGCP, Annual Report 2024, page 22); the uniform shape is an approximation applied here, not IGCP's redemption schedule. The slow and fast scenarios carry no external source and exist to bracket the assumption. The debt ratio and nominal GDP are held fixed across the horizon. Source: author calculations from Eurostat data.

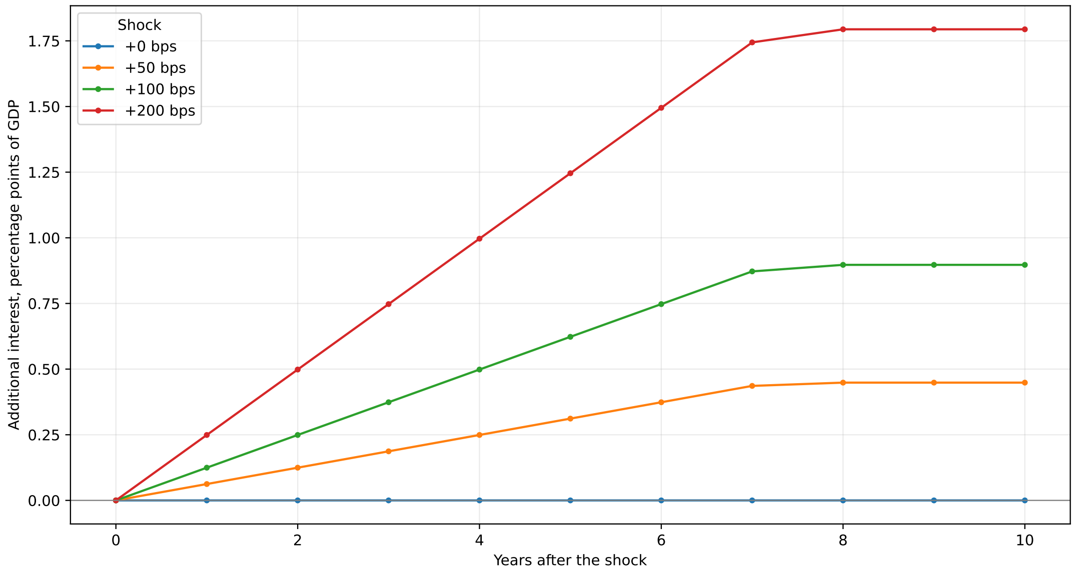
Assumptions and their provenance. Table 10 lists every input. One of them is sourced: the average maturity of the Portuguese debt stock was 7.2 years in 2024 including official loans (IGCP, *Annual Report 2024*, page 22). The central scenario converts that figure into a uniform annual repricing share of 1/7.2. The uniform shape is an approximation applied here and is *not* IGCP's redemption schedule, which is lumpy. The slow and fast scenarios carry no external source at all; they bracket the central case so the reader can see how much the result depends on the assumption rather than on the data. The debt ratio and nominal GDP are held fixed across the horizon, which isolates the repricing arithmetic. This is a stylised model and not a forecast.



Source: author calculations from Eurostat data. Stylised cohort model; not a forecast.

Figure 7: Cumulative share of the original stock repriced, by scenario.

Figure 8 reports the quantity that matters: the interest burden under each shock *minus* the burden under a zero shock. Plotting total burden paths would confound the shock with the starting level. The zero-shock case is flat at zero by construction, and each shocked path converges on the static bound in Table 9 once the whole stock has been repriced.



Source: author calculations from Eurostat data. Incremental over the zero-shock baseline; central scenario; stylised cohort model, not a forecast.

Figure 8: Additional interest relative to the zero-shock baseline, central scenario.

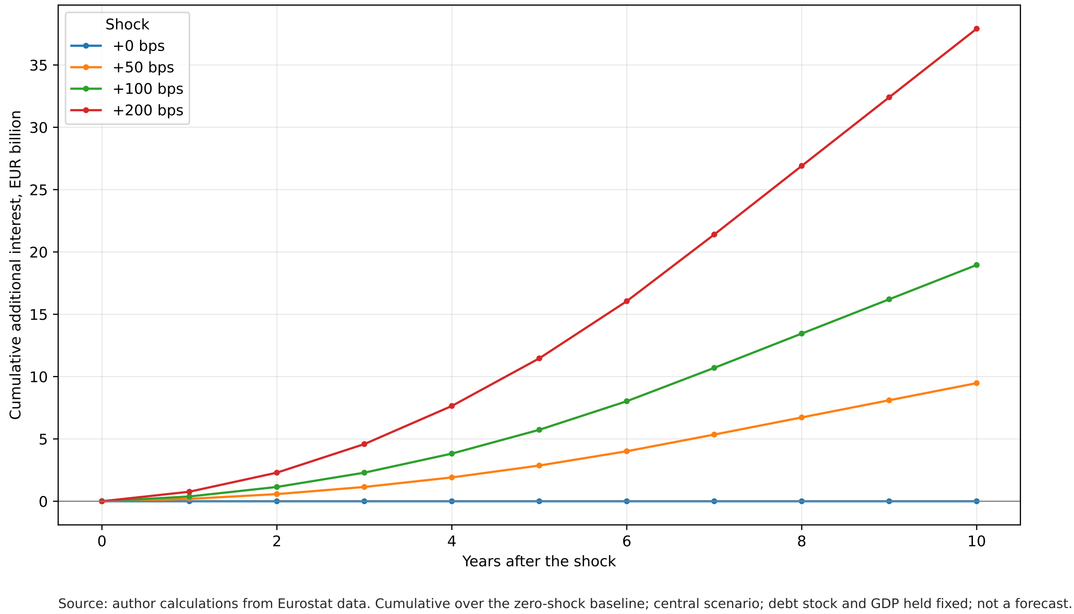


Figure 9: Cumulative additional interest in euro relative to the zero-shock baseline.

The exercise has two fiscal implications. First, maturity structure is a form of risk management: it determines how quickly market conditions enter the budget, and a longer average maturity converts a rate shock into a slow, partial repricing. Second, favourable recent debt arithmetic should not be extrapolated mechanically. If higher market yields persist, the average rate rises gradually even without an immediate fiscal shock.

7 Related Literature and Contribution

This paper is a measurement exercise, and it sits inside an accounting and empirical literature that it does not attempt to extend theoretically. The review below is organised by the questions the paper actually asks.

Debt-dynamics accounting. The arithmetic used in Section 2 is standard. Escolano [3] sets out the identity linking the change in the debt ratio to the interest-growth differential, the primary balance, and the residual stock-flow adjustment, and is explicit that the effective interest rate entering the identity is interest accrued over the *inherited* stock. That distinction is the reason this paper carries two effective-rate concepts rather than one.

The interest-growth differential. Whether a negative differential makes debt comfortable is contested. Blanchard [4] argues that when safe rates are persistently below growth, debt rollovers may carry no fiscal cost, and that this configuration is closer to the historical norm than to an exception. Mauro and Zhou [5] examine effective borrowing costs for 55 countries over up to two centuries and are more cautious: negative differentials have historically offered limited protection, because the marginal cost of borrowing can move sharply even when the average cost is low. Portugal's record over 2015–2025, where a favourable differential coexisted with a sizeable stock-flow adjustment, illustrates why the average and the margin have to be reported separately.

Fiscal sustainability. Bohn [6] proposes a sustainability test that does not require an assumption about the interest rate, asking instead whether the primary balance responds systematically to the debt ratio. This paper does not estimate such a response. It reports the debt-stabilising primary balance implied by the identity, which is a weaker and purely arithmetic object, and says so.

Euro-area sovereign borrowing costs. De Grauwe and Ji [7] test whether euro-area sovereign spreads moved beyond what fundamentals justified, and find evidence consistent with self-fulfilling liquidity dynamics for members of a monetary union that issue in a currency they do not control. This matters for the comparison in Section 6: the effective rate on an existing stock is a slow-moving average and does not carry the information that a marginal spread carries.

Debt maturity and refinancing pass-through. Missale and Blanchard [8] formalise how maturity structure governs the sensitivity of the real debt burden to shocks, introducing an effective-maturity notion. Greenwood, Hanson and Stein [9] treat maturity choice as a trade-off faced by the sovereign rather than a technical detail. The cohort model in Section 6 is a stylised version of the mechanism those papers describe, calibrated to the published average maturity of the Portuguese stock rather than estimated.

Inflation and nominal growth. Aizenman and Marion [10] assess how much of a debt ratio inflation can plausibly erode, and stress that the answer depends on maturity and on whether the inflation is anticipated. Portugal's 2022–2023 episode, where nominal growth of roughly twelve and eleven percent coincided with a sharp fall in the debt ratio, is a case where the denominator did most of the work.

Stock-flow adjustments. Weber [11] documents that the discrepancy between the change in debt and the deficit is large, persistent, and only partly explained by balance-sheet effects, and that it is smaller in more fiscally transparent countries. Seiferling [12] re-examines the same discrepancy against the government's integrated balance sheet. Both are reasons to report the stock-flow adjustment as a residual to be investigated rather than a behavioural term to be interpreted.

Portuguese debt management. The Portuguese treasury and debt agency publishes the maturity profile and issuance record used to calibrate the refinancing scenarios [13]. The average maturity of the debt stock, 7.2 years in 2024 including official loans, is the single external assumption this paper takes from outside Eurostat.

Contribution. The paper's contribution is measurement discipline rather than a new estimate. Five things are done deliberately. First, every reported quantity is generated from an auditable ESA-consistent pipeline, with provenance, observation status, and source checksums travelling with each value. Second, the marginal yield and the realised average financing cost are kept apart, and the debt-dynamics rate is distinguished from the descriptive average-debt rate, because feeding the wrong one into an identity breaks it silently. Third, the change in the burden is decomposed exactly into a financing cost effect and a debt exposure effect, with the identity checked numerically before it is interpreted. Fourth, the euro-area comparison states its eligibility rule, its denominator, its ranking convention, and the observation status of every geography, so a rank cannot be quoted without its base. Fifth, the refinancing pass-through is presented as an explicitly stylised cohort model whose assumptions are tabulated and whose only externally sourced input is named. None of this is novel economics. It is the part that is usually left implicit.

8 Discussion and Limitations

The evidence indicates that no single variable explains the full decline in Portugal's interest burden. The long-run decline from 1995 to 2025 reflects:

- a large fall in the effective interest rate from high pre-euro and convergence-period levels;
- monetary-union and later ECB-era financing conditions that lowered the average cost of the outstanding stock;
- nominal GDP growth, especially after the pandemic;
- a large decline in the debt ratio after the 2020 peak;
- improved primary balances in several recent years.

The crisis period shows the opposite combination: high debt, weak or negative nominal growth, and elevated financing stress. The recent period shows why high nominal growth can quickly reduce debt-ratio pressure, but also why interest in euros can rise even when the ratio is stable.

The paper does not prove that Portugal's fiscal position is structurally safe. It does not estimate future growth, future primary balances, or future market risk premia. It also does not model contingent liabilities, ageing costs, or debt-maturity composition in full detail. Instead, it provides the measurement and accounting basis required before those questions can be studied rigorously.

The contribution of this study is primarily empirical and reproducible. It does not propose a new macroeconomic theory. It provides a clean, auditable, annually indexed dataset that separates national-accounts interest flows, debt stocks, growth, balances, market yields, and comparator-country metrics. It also embeds validation checks that reduce the probability of silent data corruption. For policy analysis, this is a necessary foundation: fiscal interpretation is only as strong as the measurement layer underneath it.

Five limitations are material.

First, ESA interest payable is not the same as central-government cash interest. Users interested in budget execution, Treasury cash management, or debt-office operations would need a separate reconciliation layer.

Third, the average-debt interest rate is an average-cost measure, not a marginal cost of funds. It is useful for descriptive average-cost analysis but cannot replace the debt-dynamics rate or an issuance yield in debt-management analysis.

Fourth, the cross-country panel is harmonised by Eurostat definitions but does not control for maturity structure, inflation-linked debt, cash buffers, official-sector loans, or country-specific fiscal institutions.

Fifth, rate-shock simulations are deterministic arithmetic exercises. They omit feedback through growth, inflation, primary balances, market risk premia, and policy responses.

9 Conclusion

Portugal's general-government interest burden has declined dramatically in the harmonised ESA 2010 sample. The burden fell from 5.50 percent of GDP in 1995 to 1.90 percent in 2025. The nominal interest bill, however, remained material: EUR 5.96 billion in 2025. This distinction is the core empirical lesson of the paper. The euro amount, the GDP ratio, the debt ratio, the effective interest rate, and the market yield are related but distinct objects.

The sovereign-debt crisis and adjustment period remains the highest-burden regime in the sample. The post-2015 decline reflects a lower average cost of debt, refinancing effects, primary-balance improvement, and later nominal GDP growth and debt-ratio reduction. The generated endpoint decomposition shows that from 2014 to 2025 the reconstructed interest burden changed by -2.87 percentage points: the average financing-cost effect contributed -1.64 percentage points and the debt-exposure effect contributed -1.22 percentage points. Over 1996–2025, the corresponding total change was -2.88 percentage points, with -4.31 percentage points from the average financing-cost effect and 1.42 percentage points from debt exposure. The recent tightening period has not yet translated one-for-one into the interest burden because the debt stock reprices gradually and because the GDP denominator grew strongly. This does not remove the risk from sustained higher rates; it clarifies the channel through which that risk reaches the budget.

In the 2025 comparator panel, Portugal occupied an intermediate position among euro-area countries. It had a lower interest burden than Italy, Greece, Spain, Belgium, and France, but a higher burden than most other euro-area members. That ranking should be read as a harmonised descriptive comparison, not as a full fiscal-sustainability judgement.

A Appendix A: Annual ESA 2010 Portugal Table

Table 11: Annual Portugal ESA 2010 analytical table

Year	Int./GDP (	Interest	Debt/GDP	Avg. debt rate	Overall bal.	Primary bal.	Nom. growth
1995	5.5	5038.9	62.2		-5.2	0.3	
1996	4.8	4666.1	63.3	7.93	-4.7	0.1	6.16
1997	3.8	3958.3	58.7	6.56	-3.7	0.1	6.91
1998	3.1	3475.4	55.6	5.72	-4.4	-1.3	7.14
1999	2.9	3523.0	55.4	5.50	-3.0	-0.1	8.06
2000	3.0	3861.9	54.2	5.69	-3.2	-0.2	7.37
2001	3.0	4057.4	57.4	5.50	-4.7	-1.7	5.73
2002	2.8	4042.5	60.0	4.95	-3.3	-0.5	4.99
2003	2.7	3883.1	63.9	4.34	-5.7	-3.0	2.46
2004	2.6	3908.8	67.1	4.00	-6.2	-3.6	4.23
2005	2.6	4071.3	72.2	3.76	-6.1	-3.5	4.14
2006	2.8	4626.2	73.7	3.90	-4.2	-1.4	4.86
2007	3.0	5205.1	72.7	4.16	-2.9	0.1	5.55
2008	3.1	5572.8	75.5	4.24	-3.8	-0.7	2.06
2009	3.0	5230.1	87.6	3.62	-9.9	-6.9	-2.06
2010	2.9	5245.5	99.9	3.15	-11.4	-8.5	2.53
2011	4.3	7521.6	114.0	3.95	-7.7	-3.4	-1.97
2012	4.8	8136.9	128.6	3.90	-6.2	-1.4	-4.41
2013	4.8	8139.3	130.8	3.70	-5.2	-0.4	1.27
2014	4.8	8334.4	132.5	3.68	-7.4	-2.6	1.47
2015	4.5	8132.8	131.0	3.50	-4.5	0.0	3.58
2016	4.1	7633.3	131.2	3.18	-1.9	2.2	3.90
2017	3.7	7299.4	126.0	2.97	-3.0	0.7	4.90
2018	3.3	6805.8	121.1	2.75	-0.4	2.9	4.85
2019	2.9	6226.9	116.1	2.50	0.1	3.0	4.63
2020	2.8	5697.1	134.1	2.20	-5.8	-3.0	-6.27
2021	2.4	5117.8	123.9	1.90	-2.8	-0.4	7.69
2022	1.9	4608.0	111.2	1.71	-0.3	1.6	12.69
2023	2.1	5553.3	96.9	2.08	1.1	3.2	10.82
2024	2.0	5934.8	93.5	2.23	0.6	2.6	7.19
2025	1.9	5964.5	89.7	2.18	0.7	2.6	5.85

B Appendix B: Variable Definitions

Table 12: Core analytical variables

Column	Unit	Definition
`year`	year	Calendar year.
`interest_mio_eur`	EUR million	General-government interest payable.
`interest_pct_gdp`	percent of GDP	Preferred interest burden: official Eurostat ratio with calculated fallback.
`government_expenditure_mio_eur`	EUR million	General-government total expenditure.
`government_expenditure_pct_gdp`	percent of GDP	Preferred total-expenditure ratio: official Eurostat ratio with calculated fallback.
`government_revenue_mio_eur`	EUR million	General-government total revenue.
`government_revenue_pct_gdp`	percent of GDP	Preferred total-revenue ratio: official Eurostat ratio with calculated fallback.
`nominal_gdp_mio_eur`	EUR million	GDP at current market prices.
`debt_mio_eur`	EUR million	Maastricht consolidated gross debt.
`debt_pct_gdp`	percent of GDP	Preferred debt ratio.
`average_debt_`		

Column	Unit	Definition
interest_rate_pct	percent	Interest payable divided by average debt.
debt_dynamics_ interest_rate_pct	percent	Interest payable divided by previous-year debt.
overall_balance_pct_gdp	percent of GDP	Net lending or borrowing.
primary_balance_pct_gdp	percent of GDP	Overall balance plus interest expenditure.
nominal_gdp_growth_pct	percent	Annual growth of current-price GDP.
real_gdp_growth_pct	percent	Chain-linked volume growth rate.
gdp_deflator_growth_pct	percent	Derived GDP deflator growth.
interest_growth_ differential_pct	percentage points	Debt-dynamics interest rate minus nominal GDP growth.
debt_stabilising_primary_ balance_before_sfa_pct_gdp	percent of GDP	Primary balance required to stabilise debt absent stock-flow adjustment.
stock_flow_adjustment_pp	percentage points	Debt-dynamics residual.
ten_year_yield_pct	percent	Long-term government bond yield.
source	category	Source family; Eurostat in this build.
accounting_basis	category	Accounting framework label.
observation_status	category	Observed or forecast status.
retrieval_timestamp_utc	timestamp	Joined UTC source retrieval timestamp where available.
source_checksum_sha256	hash	Joined source checksum metadata.

C Appendix C: Source and Validation Notes

Eurostat source tables are `gov_10a_main`, `gov_10dd_edpt1`, `nama_10_gdp`, and `irt_lt_mcby_a`. The latest validation result passed all error-severity checks (yes), with 0 error-level and 1 warning-level failures. Source data were retrieved between 20260728T200454Z and 20260728T200500Z. The observation statuses accepted into the euro-area comparison are observed, provisional. The largest official-versus-reconstructed ratio discrepancy is 1.0152 percentage points, in 1997, 1998; the per-year rows for those years appear above. A data-revision log comparing the current processed vintage with the previous one is written to `reports/data_revision_log.csv` and `reports/data_revision_log.md`. An error-level validation failure stops the report build.

D Appendix D: Replication Materials

The replication workflow produces the following analytical outputs:

- `data/processed/portugal_debt_interest.csv`;
- `data/processed/portugal_debt_interest.sqlite`;
- `data/processed/eurostat_panel_metrics.csv`;
- `reports/validation.json`;
- `reports/summary.md`;
- publication-oriented figures in `reports/figures/`.

Raw source downloads include retrieval timestamps, request metadata, payload sizes, and checksums. Processed rows preserve source-vintage and checksum metadata so revisions can be audited.

E Appendix E: Reproducibility

The full command sequence is documented in the repository README. The pipeline is reproduced with `pt-debt all --config config/default.yaml`, which performs acquisition, the analytical build, validation, figure and table generation, and the Markdown summary.

References

[1] Eurostat. *European System of Accounts: ESA 2010*. Publications Office of the European Union.

[2] Eurostat. *Government finance statistics: revenue, expenditure, deficit, debt, GDP, and long-term interest rate datasets*. Eurostat database.

[3] Escolano, J. (2010). *A Practical Guide to Public Debt Dynamics, Fiscal Sustainability, and Cyclical Adjustment of Budgetary Aggregates*. International Monetary Fund, Technical Notes and Manuals 10/02. Washington, DC: IMF. https://www.imf.org/external/pubs/ft/tnm/2010/tnm1002.pdf

[4] Blanchard, O. (2019). Public Debt and Low Interest Rates. *American Economic Review*, 109(4), 1197–1229. DOI: 10.1257/aer.109.4.1197

[5] Mauro, P., & Zhou, J. (2021). $r - g < 0$: Can We Sleep More Soundly? *IMF Economic Review*, 69(1), 197–229. DOI: 10.1057/s41308-020-00128-y

[6] Bohn, H. (1998). The Behavior of U.S. Public Debt and Deficits. *Quarterly Journal of Economics*, 113(3), 949–963.

[7] De Grauwe, P., & Ji, Y. (2013). Self-fulfilling crises in the Eurozone: An empirical test. *Journal of International Money and Finance*, 34, 15–36. DOI: 10.1016/j.jimonfin.2012.11.003

[8] Missale, A., & Blanchard, O. J. (1994). The Debt Burden and Debt Maturity. *American Economic Review*, 84(1), 309–319.

[9] Greenwood, R., Hanson, S. G., & Stein, J. C. (2015). A Comparative-Advantage Approach to Government Debt Maturity. *The Journal of Finance*, 70(4), 1683–1722. DOI: 10.1111/jofi.12253

[10] Aizenman, J., & Marion, N. (2011). Using inflation to erode the US public debt. *Journal of Macroeconomics*, 33(4), 524–541.

[11] Weber, A. (2012). *Stock-Flow Adjustments and Fiscal Transparency: A Cross-Country Comparison*. International Monetary Fund, Working Paper 12/39. Washington, DC: IMF.

[12] Seiferling, M. (2013). *Stock-Flow Adjustments, Government's Integrated Balance Sheet and Fiscal Transparency*. International Monetary Fund, Working Paper 13/63. Washington, DC: IMF.

[13] Agência de Gestão da Tesouraria e da Dívida Pública (IGCP) (2025). *Annual Report 2024*. Lisbon: IGCP. https://www.igcp.pt/sites/default/files/2025-11/RA2024_20251118.pdf